

L'ASSOMPTION, QC, Canada

WMO: 715240

Lat: 45.8094N

Lon: 73.4347W

Elev: 69

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.2	-8.7	-22.9	1.6	-12.8	-18.0	2.1	-7.6	23.3	28.2	20.3	26.6	4.4	240	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	86.9	71.6	83.9	69.7	81.2	68.3	74.3	82.8	72.4	80.1	70.7	77.9	9.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.6	117.1	78.8	69.9	110.1	76.8	68.2	103.9	75.2	37.9	82.5	36.2	80.2	34.7	77.9	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.5	16.1	14.0	DB	-21.8	91.4	6.1	2.7	-26.2	93.4	-29.8	95.0	-33.2	96.5	-37.6	98.5	(4)
			WB	-22.3	77.8	6.1	2.2	-26.7	79.4	-30.3	80.7	-33.7	82.0	-38.1	83.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	13.4	16.4	27.0	42.1	56.1	65.2	69.4	67.6	60.0	47.1	34.7	21.6	(6)
(7)		DBStd	21.61	13.14	11.35	10.77	8.14	7.71	6.67	5.26	5.45	7.49	7.84	8.86	11.60	(7)
(8)		HDD50	4577	1133	941	714	260	26	0	0	7	153	462	881		(8)
(9)		HDD65	8247	1598	1361	1178	688	292	78	17	36	187	557	909	1346	(9)
(10)		CDD50	2214	0	0	1	22	216	455	603	547	305	62	3	0	(10)
(11)		CDD65	410	0	0	0	1	17	83	155	117	36	1	0	0	(11)
(12)		CDH74	3343	0	0	1	14	219	735	1239	860	268	7	0	0	(12)
(13)		CDH80	837	0	0	0	2	52	204	327	192	60	0	0	0	(13)
(14)	Wind	WSAvg	6.3	6.7	6.8	7.0	7.4	6.8	6.0	5.5	5.2	5.5	6.1	6.5	6.5	(14)
(15)	Precipitation	PrecAvg	40.0	2.9	2.2	2.5	3.3	3.5	3.9	3.8	4.1	3.5	3.8	3.1	3.5	(15)
(16)		PrecMax	48.3	5.6	4.6	5.2	6.1	7.4	8.8	6.5	7.2	6.8	6.8	5.6	6.6	(16)
(17)		PrecMin	30.9	0.9	1.0	0.7	0.4	1.9	1.1	2.4	1.9	1.1	1.1	0.5	0.7	(17)
(18)		PrecStd	4.6	1.2	0.9	1.1	1.5	1.3	1.8	1.0	1.2	1.4	1.5	1.1	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	42.3	57.4	74.9	86.0	89.7	91.1	88.4	86.7	74.2	61.3	48.3	(19)
MCWB			44.1	38.3	47.7	59.1	67.9	72.4	74.9	72.3	71.5	62.8	56.7	45.1	(20)	
2%		DB	39.2	38.8	48.7	66.9	79.9	85.5	86.7	84.7	80.6	68.3	55.6	42.7	(21)	
		MCWB	36.8	35.9	41.7	52.8	62.9	69.6	72.4	70.6	68.9	59.4	51.3	40.7	(22)	
5%		DB	36.0	36.2	43.8	60.9	75.2	81.8	83.5	81.8	76.5	63.2	51.2	38.9	(23)	
		MCWB	34.3	33.9	38.5	49.6	60.1	67.7	70.5	69.2	66.6	56.2	47.0	36.8	(24)	
10%		DB	32.6	33.6	40.5	55.8	70.6	78.1	80.7	78.9	72.8	59.3	47.1	35.9	(25)	
		MCWB	31.2	31.8	36.2	46.4	58.3	65.2	68.9	67.8	64.4	53.6	43.5	34.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.1	39.5	50.4	61.5	70.5	75.4	77.6	75.9	73.6	65.2	57.8	46.1	(27)
MCDWB			45.1	40.9	55.2	71.5	81.4	85.6	87.7	84.0	82.0	71.4	59.8	47.6	(28)	
2%		WB	37.5	36.3	42.6	55.0	66.4	72.4	75.0	73.3	70.6	60.3	52.1	40.9	(29)	
		MCDWB	38.4	38.1	46.9	62.5	74.8	81.6	83.1	81.2	77.2	66.3	55.3	42.5	(30)	
5%		WB	34.3	34.2	39.3	50.5	63.2	70.0	72.8	71.5	68.3	56.9	47.4	37.1	(31)	
		MCDWB	35.8	35.9	42.8	59.0	71.3	77.7	79.9	78.4	74.8	61.8	50.5	38.6	(32)	
10%		WB	31.3	32.0	36.7	47.3	59.9	67.8	71.0	69.8	65.8	54.2	44.0	34.2	(33)	
		MCDWB	32.7	33.7	39.9	54.5	67.5	75.3	78.1	76.3	71.6	58.1	46.7	35.6	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.4	17.6	16.5	18.1	19.8	19.2	19.1	19.4	19.7	16.2	13.2	13.6	(35)
MCDBR			16.6	15.5	18.8	28.7	27.2	24.2	22.0	22.1	22.8	22.1	19.0	13.6	(36)	
MCWBR			16.0	13.9	13.5	16.8	13.8	11.7	10.8	10.9	12.8	14.3	15.5	12.4	(37)	
5% WB		MCDWR	16.1	14.8	17.1	24.1	22.6	20.3	19.0	18.9	20.2	19.5	17.8	13.2	(38)	
		MCWBR	15.7	13.9	13.2	16.0	13.8	11.5	10.7	10.5	12.3	13.9	15.6	12.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.255	0.289	0.335	0.364	0.392	0.437	0.428	0.415	0.376	0.337	0.307	0.293	(40)	
(41)		taud	2.347	2.238	2.241	2.341	2.328	2.209	2.240	2.275	2.372	2.486	2.481	2.290	(41)	
(42)		Ebn at Noon	274	281	282	284	278	265	266	264	266	263	252	239	(42)	
(43)		Edh at Noon	23	32	37	37	39	44	43	39	33	25	21	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	456	751	1150	1397	1624	1784	1782	1571	1240	718	432	313	(44)	
(45)		RadStd	55	80	97	155	159	133	141	73	102	61	56	27	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	(47)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137	N/A
Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page